Shashikiran Javvaji

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PROFILE

Principal/Staff-level software engineer with 18+ years of experience designing, building, and scaling distributed systems across embedded platforms, cloud infrastructure, and backend services. Currently a Principal Engineer in Charter's Emerging Technology group, driving 0-to-1 proof-of-concepts in latency measurement, data monetization, and network-as-a-service APIs on AWS. Previously worked as a Staff Software Engineer at EchoStar/Dish, where large-scale backend services, telemetry pipelines, and patented product features were delivered for millions of users. Recently focused on integrating AI-native and agentic tools (Kiro, Cursor, Claude Code, AWS Bedrock) into the development lifecycle to accelerate design, implementation, and experimentation.

SKILLS

Languages: C, Java, Python, Shell Scripting
AI & Agentic Development: AWS Bedrock, LangChain, MCP, Kiro, Cursor, Claude Code, Copilot
Cloud Technology: AWS - Bedrock, AgentCore, EC2, Lambda, IAM, MSK, Kinesis, Firehose, S3, Kubernetes
Web and Database Technologies: JSON, Html, MySQL, MongoDB
Data and Observability: ELK(Elasticsearch Logstash Kibana)***,*** Grafana, OpenSearch, Splunk, Tableau
Simulators: Eclipse, Android Studio, IntelliJ, Atom, Visual Studio
Operating Systems: Linux, Mac, Android, Windows, iOS, OpenWRT, RDK
Packages: *GIT*, Argo, ClearCase, Photoshop, Flash, Atlassian, Plume, Docker, OpenSync, Agora, NumPy, Pyshark, NetScout
Project Management: Agile, Scaled Agile, Waterfall Programming
Communication Protocols: TCP/IP, UDP, HTTP, MQTT, TR-069, WebSocket, DHCP, DNS

EXPERIENCE

06/23 – Present
Denver, CO

Director/Principal Engineer II
Charter Communications (Emerging Technology)
The Emerging Technology team is a startup inside a corporate conglomerate which focuses on developing and showcasing unique customer-oriented proof of concepts(POC) that have potential to be commercialized in the larger Charter footprint.

Projects:

AI-Augmented Development: Incorporated AI-native tools (Kiro, Cursor, Claude Code, AWS Bedrock) into design and implementation workflows to generate specs, scaffold services, perform multi-file refactors, and document APIs, accelerating POC delivery for Personal Assistant and reducing time spent on boilerplate and debugging.

Latency Measurement - Led the end-to-end design, integration, and deployment of a latency measurement platform across Charter's network, including firmware coordination for field trials, third-party agent integration (Veego, Domos), cloud infrastructure setup using AWS IoT Core and MSK, development of internal analytics tools, and exploration of Plume Design data sources.

Reference Design Kit - RDK - Evaluated and integrated RDK for next-generation Charter routers by collaborating with external vendors to build Raspberry Pi-based proof of concepts(POC) and interfacing with the RDK team to assess existing solutions across multiple vendors.

Data Monetization - Unlock new, privacy-conscious insights from Charter network data to drive value for advertising partner Spectrum Reach by enabling more effective planning, purchasing, and measurement for Reach's programming and brand partners.

Bryte IQ - Development of Network as a service APIs to leverage real-time network and device data to help various industries such as e-commerce, finance, transportation, and consumer electronics reduce fraud, authenticate users, verify transactions, and enhance connectivity.

Eidolon - Developed a video conferencing application featuring digital humans generated via a custom photo booth, set up Apple Vision Pro with AWS Ant Media for high-resolution video testing and latency analysis, and established a lab for L4S testing to characterize performance across various usage scenarios.

06/08 – 06/23 **Staff Software Engineer**
Denver, CO **EchoStar Satellite LLC/Dish Technologies**

Projects:

Metadata Delivery - Built delivery pipelines for Dish, Sling, and web clients using AWS EKS, Argo, and ELK for streamlined deployment and insights

Cloud Search Data - Developed voice search-ready Elasticsearch indices using AWS EKS, Kubernetes, and Docker, with CRON jobs to procure and sanitize data.

Dish Home Gateway - Collaborated with hardware and ACS teams to analyze lab setups, developing CRON jobs to procure and sanitize data, enabling Dish to better understand and utilize ACS features.

Cloud Update - Developed cloud-based backend services using Spring Boot to enable over-the-air software updates for set-top boxes, replacing satellite targeting and significantly reducing time and costs.

Real Time Reporting - Built a real-time data pipeline and Grafana dashboards for millions of deployed set-top boxes to surface status and errors, enabling early bug detection, informed deployment decisions, and proactive fixes before software reached the field.

Android TV - Led the development of AirTV Android onboarding and remote finder app in collaboration with five partners, including Google TV, enabling an innovative, cost-effective device solution for Dish.

Ipvod UI - Implemented next-generation animated UI using C on legacy and new set-top boxes with low CPU power, enabling rich media support and avoiding hardware replacements for Bell Canada.

Screen Profiler - Developed a unique screen profiling feature for EchoStar, enabling support for multiple graphical user interfaces and allowing clients in Canada to switch UI palettes without new software releases, reducing deployment costs.

UI Skin Changer - Developed a skin changer feature for set-top boxes that allowed users to switch UI themes via remote, view on-screen graphic resources, and customize UI colors for multi-customer demos without redeploying software.

Direct Call Sign Entry - Invented and implemented a patented feature (US Patent No: 8,225,356) enabling users to select channels by name instead of number

Multilingual Text Software - Developed patented methods for compacting and combining graphics using image processing and typography techniques, built multilingual UI support, maintained production codebases for Dish, Bell, and TELUS, and implemented end-to-end solutions on Linux using Slick-Edit.

ENGINEERING LEADERSHIP

- Mentored interns and junior engineers for over 15 years across organizations including Charter Communications and Dish Network, helping them grow technically and professionally across projects involving embedded systems, firmware, and cloud infrastructure.
- At Charter, provided hands-on mentorship for projects involving real-time network telemetry, latency measurement, and integration with analytics tools
- At Dish Network, guided junior engineers in developing set-top box features, image processing enhancements, and multilingual UI implementations. Helped new engineers understand the architecture of satellite communication systems and test automation practices.
- Led by example in cross-functional collaboration mentoring engineers working across firmware, DevOps, and cloud services. Helped them interface effectively with platform, QA, and product teams to deliver reliable software solutions.
- Provided code reviews, architectural guidance, and documentation best practices across multiple platforms, fostering technical rigor and autonomy in mentees.
- Built a culture of technical ownership by encouraging problem-solving, constructive feedback, and continuous improvementin engineering quality.

EDUCATION

MSEE, The University of the Texas at El Paso (UTEP), 05/08; GPA: 3.89/4.0
Machine Learning Certificate, Cornell University, Credential - kLKseZRMBc
Technology Leadership Certificate, Cornell University, Credential - EfQpCRdVwE

References Available on Request